



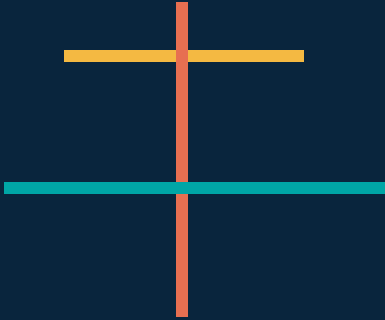
TTC Subway Delay Analytics

The Toronto Transit Commission (TTC) subway system is a critical part of Toronto's public transportation network, servicing over a million riders daily across multiple lines.

The subway delays have been a persistent issue, impacting the efficiency and reliability of the service. Analyzing the delay patterns will give insights to improve the service.

Toronto subway service-delay dataset

Prepared for an analytics portfolio by Group 2



Project purpose

Turn operational delay records into a clear view of where, when, and why service reliability changes.

This project analyzes historical TTC subway operational and delay data spanning **January 2022 to June 2026 (4.5 years)** to identify core service patterns, evaluate transit reliability across routes, and diagnose long-term drivers of service disruptions.

A delay event can create impacts that last beyond its reported minutes. Riders may wait through a wider train gap, miss a transfer, arrive late, or change travel plans.

How delays affect commuters

- **Delay event**

A train or station issue changes normal service.

- **Wider headway**

The gap between trains can increase.

- **Rider consequence**

Crowding, missed connections, and uncertain arrival times can follow.

- **Planning signal**

Repeated patterns show where clearer alerts or resilience work may help.

Questions the Analysis should Answer

01

What are the delay pattern can be derived from data across multiple years?

02

Which stations and lines accumulate the most delay minutes?

03

Which causes occur often, and which causes create the longest disruptions?

04

When do delays cluster by hour, day, month, or weather season?

Visualize the delay patterns

Time patterns

Heatmap of delay minutes by hour and day of week. Monthly trend for seasonality and emerging changes.

Location patterns

Rank stations and lines by total delay minutes, event count, and median delay. Map the highest-burden locations.

Cause patterns

Compare codes by frequency versus total delay minutes. Highlight causes that have a low count but high severity.

Service patterns

Compare direction, line, and vehicle where data quality permits. Look for recurring combinations rather than single events.

Multiple Excel and CSV file with TTC subway operational and delay data

Combined dataset: 109,967 rows x 10 columns

Dictionary table with Delay Code and corresponding description



Data Source: [TTC Subway Delay Data - Open Data Toronto](#)

TTC Subway Delay Dataset and fields

Each row represents a reported subway delay event. The fields support time, location, cause, and impact analysis.

Event identity	Time	Location	Cause	Impact	Equipment
_id Unique record identifier	• Date • Time When the event occurred	• Station • Bound • Line Where service was affected	Code What is the delay type	• Min Delay • Min Gap How many minutes of delay in Service and headway effect	Vehicle What is the Associated train or vehicle number

Portfolio storyline and deliverables

The portfolio should show a complete analytics workflow, from raw public data to a decision-ready dashboard.

01

Data profiling

Check missing values, duplicate events, code coverage, time parsing, and outliers in delay and gap minutes.

02

Data preparation

Create timestamp, month, hour, cause group, and severity fields. Retain original records for auditability.

03

Exploratory analysis

Identify trends, stations, lines, and codes that dominate frequency or delay burden.

04

Dashboard

Interactive data visualization dashboard that consolidates key metrics and insights into a single view, and visualize identified trends and patterns

Data Cleaning

- Bound column has more null rows and duplicate information so dropped the column.
- Many typos in the source data, replaced the same with the correct spelling.
- Line 3 / SRT → train service ended July 24, 2023. So, we no longer needed the corresponding data for analysis. Dropping the Line 3/SRT rows from data frame.
- Normalizing the Line column: Multiple variants of the same line are causing inconsistency in data. (eg. YU, YUS, Line 1m etc.)
- Found some invalid names in Station column. Removed them from dataset.
- Some 'Line' data is missing but can be retrieved by mapping

Columns	Null rows
Date	0
Time	0
Day	0
Station	0
Code	0
Min Delay	0
Min Gap	0
Bound	37771
Line	274
Vehicle	0

Data Wrangling

Fill Missing Line Values

- Count missing values in the Line column.
- Use the station-to-line mapping to fill missing Line values.
- Preserve existing non-null Line values.
- Remove rows where the Line value remains unresolved.
- Create a clean dataset for further analysis and visualization.

Delay code to Category mapping

- Category mapping for TTC delay codes into 7 high-level cause categories (Code_Category) based on Code description.
- This reduced complexity while preserving the underlying reason for each delay

Code_Category	Codes count
Infrastructure & Signals	31
Train & Equipment	24
Other	22
Operations & Workforce	19
Safety & Security	13
Environmental & Cleanliness	12
Passenger & Medical	8

Data Wrangling

Creating Date and Time Features

The Datetime column is used to extract and create additional date and time features for analyzing TTC delay patterns.

The following features are created:

- **Year** – Calendar year of the delay.
- **Month** – Numeric month of the delay.
- **Day_Num** – Day of the month.
- **Hour** – Hour of the day, used for time-based analysis.
- **Weekday** – Numeric day of the week, where Monday = 0 and Sunday = 6.
- **DayName** – Name of the day of the week.
- **Peak_Hour** – Binary indicator identifying TTC peak periods: 7:00–10:00 and 16:00–19:00.
- **Weekend** – Binary indicator where 1 represents Saturday/Sunday and 0 represents weekdays.
- **Season** – Categorizes each record into Winter, Spring, Summer, or Fall based on the month.

These features support analysis of TTC delays by **year, month, day, hour, weekday/weekend, peak periods, and season**, and can be used for both exploratory analysis and Power BI visualizations.